```
C:\Users\minfa\AI Test\Write-ups\TokenAwakening>python eval_prompts.py
C:\Users\minfa\AI Test\Write-ups\TokenAwakening\eval_prompts.py:11:
DeprecationWarning: 'LLMTestCaseParams' is deprecated and will be removed
in a future release. Use 'SingleTurnParams' instead.
  from deepeval.test_case import LLMTestCase, LLMTestCaseParams
Extracting text from
C:\Users\minfa\Downloads\2025_Federal_Reserve_Stress_Test_Results.pdf...
Extracting text from
C:\Users\minfa\Downloads\2026_Federal_Reserve_Stress_Test_Results.pdf...
Extracting text from
C:\Users\minfa\Downloads\2025_Federal_Reserve_Stress_Test_Scenario.pdf...
Extracting text from
C:\Users\minfa\Downloads\2026_Federal_Reserve_Stress_Test_Scenario.pdf...
Extraction complete. Read 503586 characters.


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[Output]
## Executive Summary

This analysis attributes the year-over-year (YoY) changes in Citigroup's
projected loan loss rates for its Credit Cards and Commercial Real Estate
(CRE) domestic portfolios under the Federal Reserve's Severely Adverse
Stress Test Scenarios for 2025 and 2026.

Citigroup's projected loan loss rate for **Credit Cards decreased by 0.3
percentage points** from 16.6 percent in 2025 to 16.3 percent in 2026.
This modest improvement in the credit card portfolio's performance under
stress can be primarily attributed to a **less severe decline in real
disposable income growth** in the 2026 scenario compared to 2025 (trough
of -1.1% in 2026 vs. -6.0% in 2025). This indicates a less pronounced
impact on consumers' ability to repay, which likely offset the slightly
more deflationary environment (CPI inflation trough of 1.0% in 2026 vs.
1.3% in 2025). The peak unemployment rate, a key driver for consumer
credit, remained consistent at 10.0 percent across both scenarios,
suggesting no additional stress from this factor.

Conversely, Citigroup's projected loan loss rate for **Commercial Real
Estate, domestic, increased significantly by 2.6 percentage points** from
8.3 percent in 2025 to 10.9 percent in 2026. This deterioration is
directly linked to a **more severe decline in commercial real estate
prices** (a 39 percent decline in 2026 compared to a 30 percent decline
in 2025) and a **higher peak in BBB corporate yields** (8.2 percent in
2026 versus 6.0 percent in 2025). These factors indicate increased stress
on collateral values and higher financing costs for corporate borrowers,
respectively, which are critical for CRE portfolio performance. While the
overall real GDP decline was less severe in 2026 (-4.6% vs. -7.8% in
2025), and the peak unemployment rate remained unchanged, the direct and
intensified stress on CRE prices and corporate financing conditions
outweighed these mitigating factors, leading to a higher projected loss
rate for the portfolio. The 2026 stress test results explicitly note that
```

"more severe changes in corporate bond spreads and commercial real estate prices" contributed to greater loan losses year-over-year.

In summary, while the 2026 severely adverse scenario presented a slightly less challenging environment for consumer income, leading to a marginal improvement in credit card loss rates, it introduced significantly more severe stress to commercial real estate valuations and corporate financing, resulting in a notable increase in CRE domestic loan loss rates for Citigroup.

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1. Loss Rate Change

Citigroup's Projected Loan Loss Rates (Severely Adverse Scenario)

Portfolio	2025 Loss Rate (Percent)	2026 Loss Rate (Percent)	YoY Change (Percentage Points)
Credit cards	16.6	16.3	-0.3
Commercial real estate, domestic	8.3	10.9	+2.6

Source: 2025 Federal Reserve Stress Test Results, Table 10; 2026 Federal Reserve Stress Test Results, Table 10.

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2. Macroeconomic Drivers

Based on the provided documents and financial domain knowledge:

*   **Credit Cards:**
    *   **Unemployment Rate:** Directly impacts consumers' ability to repay.
    *   **Real Disposable Income Growth:** Affects consumer spending power and capacity to service debt.
    *   **CPI Inflation Rate:** Influences purchasing power and the real burden of debt.
    *   *Supporting qualitative commentary (general for loan portfolios):* "Some loan portfolios are sensitive to home prices or unemployment rates and may experience high stressed loss rates due to the considerable stress on these factors in the severely adverse scenario." (2025 Federal Reserve Stress Test Results, p. 21)

*   **Commercial Real Estate (CRE) Domestic:**
    *   **Commercial Real Estate Prices:** Directly impacts the value of collateral and potential losses upon default.
    *   **Unemployment Rate:** Affects demand for commercial space and tenant solvency.
    *   **Real GDP Growth:** Reflects overall economic activity, influencing business expansion and demand for commercial properties.

    *   **BBB Corporate Yield:** Indicates the cost of corporate financing and the health of corporate borrowers, impacting their ability to manage CRE investments and debt.
    *   *Supporting qualitative commentary:* "This year, the supervisory severely adverse scenario is characterized by a severe global recession accompanied by a period of heightened stress in commercial and residential real estate markets, as well as in corporate debt markets." (2025 Stress Test Scenarios, p. 1). The 2026 results explicitly state: "Combined with more severe paths for certain scenario variables in this year’s stress test, including more severe changes in corporate bond spreads and commercial real estate prices, this led to greater loan losses year-over-year." (2026 Federal Reserve Stress Test Results, p. 3).

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3. Macroeconomic Variable Analysis

The following table details the severity of key macroeconomic variables under the Severely Adverse Scenario for 2025 and 2026, and their year-over-year differences.

Macro Variable	2025 Severity (Severely Adverse Scenario)	2026 Severity (Severely Adverse Scenario)	YoY Difference (2026 - 2025)
Unemployment Rate (Peak)	10.0% (Q3 2026)	10.0% (Q3 2027)	0.0 p.p.
Real Disposable Income Growth (Trough)	-6.0% (Q1 2025)	-1.1% (Q2 2026)	+4.9 p.p.
CPI Inflation Rate (Trough)	1.3% (Q3 2025)	1.0% (Q4 2026)	-0.3 p.p.
Commercial Real Estate Prices (Peak-to-Trough Decline)	-30%	-39%	-9.0 p.p.
Real GDP Growth (Peak-to-Trough Decline)	-7.8%	-4.6%	+3.2 p.p.
BBB Corporate Yield (Peak)	6.0% (Q3 2025)	8.2% (Q2 2026)	+2.2 p.p.

Note on Severity Metrics:
*   **Unemployment Rate:** Peak value reached during the scenario.
*   **Real Disposable Income Growth:** Trough (lowest quarterly growth rate) during the scenario.
*   **CPI Inflation Rate:** Trough (lowest quarterly inflation rate) during the scenario.

*   **Commercial Real Estate Prices:** Total percentage decline in the index level from the scenario's start to its trough, as stated in the qualitative summary of the scenarios.
*   **Real GDP Growth:** Total percentage decline in the GDP level from the scenario's start to its trough, as stated in the qualitative summary of the scenarios.
*   **BBB Corporate Yield:** Peak value reached during the scenario.

Source: 2025 Stress Test Scenarios, Table 4.A; 2026 Stress Test Scenarios, Table 4.A; 2025 Stress Test Scenarios, Executive Summary p.1, p.5; 2026 Stress Test Scenarios, Executive Summary p.4, p.13.

[Usage] Prompt Tokens: 168,661 | Output Tokens: 7,855
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      → Macro numbers found. Running DeepEval Economic Reasoning Check...
      → DeepEval Score: 90.0% | Reason: The response strongly aligns with all four evaluation steps. It accurately references macroeconomic severity changes in the executive summary (CRE prices declining 30% vs 39%, BBB yields 6.0% vs 8.2%, disposable income trough -6.0% vs -1.1%). It logically connects these changes to loss rate movements: the -0.3pp credit card improvement is tied to less severe disposable income decline, and the +2.6pp CRE increase is coherently linked to more severe CRE price declines and higher corporate yields, with correct directional and magnitude reasoning. The economic reasoning linking variables to portfolios is valid and domain-appropriate (unemployment/income for consumer credit; CRE prices/corporate yields for CRE). The synthesis avoids contradictions and explicitly acknowledges mitigating factors (less severe GDP decline, unchanged unemployment) rather than omitting them, and cites the Fed's explicit commentary on corporate bond spreads and CRE prices. Minor weakness is that the analysis relies on stated document data without any verifiability check possible here, but the internal consistency and completeness are excellent.
      → DeepEval Usage: Input 4,316 / Output 645 | Total Cost: $0.037705
✅ Validation: PASSED

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[Output]
To: Executive Committee
From: Chief Risk Officer
Date: July 1, 2026
Subject: Macroeconomic Attribution Analysis of YoY Change in Citigroup's Projected Stressed Losses

This memorandum provides a macroeconomic attribution analysis for the year-over-year (YoY) change in projected stressed loss rates for Citigroup’s Credit Card and Commercial Real Estate (CRE) portfolios, based on the 2025 and 2026 CCAR results and their underlying scenarios.

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1. Loss Rate Change

Projected loan loss rates for Citigroup under the Severely Adverse Scenario have been extracted from the 2025 and 2026 Federal Reserve Stress Test Results documents. The YoY change is as follows:

*   **Credit Cards:** The projected loss rate decreased from **16.6%** in the 2025 test to **16.3%** in the 2026 test, a change of **-0.3 percentage points**.
*   **Commercial real estate, domestic (CRE):** The projected loss rate increased from **8.3%** in the 2025 test to **10.9%** in the 2026 test, a change of **+2.6 percentage points**.

2. Macroeconomic Drivers

Based on the provided documents and standard risk management principles, the primary macroeconomic variables driving losses for these portfolios are:

*   **Credit Cards:** The **Unemployment Rate** is the principal driver, as it directly impacts consumers' ability to service unsecured debt.
*   **Commercial real estate, domestic (CRE):** Losses are driven by both the **Commercial Real Estate Price Index** and the **Unemployment Rate**, which affects tenant demand and property income.

3. Macroeconomic Variable Analysis

The severity of the key macroeconomic drivers in the Severely Adverse Scenario for each stress test year is detailed below. The 2026 scenario featured a significantly more severe decline in CRE prices compared to the 2025 scenario, while the peak unemployment rate remained constant.

Macro Variable	2025 Severity	2026 Severity	YoY Difference
Peak Unemployment Rate	10.0%	10.0%	0.0 p.p.
CRE Price Decline (Trough)	-30%	-39%	-9.0 p.p.

4. Executive Summary

The year-over-year variance in Citigroup’s projected stressed loss rates for its Credit Card and CRE portfolios is directly attributable to specific changes in the Federal Reserve’s Severely Adverse Scenarios.

The projected loss rate for the **Credit Card** portfolio remained stable, decreasing by a nominal 0.3 percentage points. This stability is explained by the consistency of its primary macroeconomic driver. The 2025 and 2026 scenarios both project a peak U.S. unemployment rate of 10.0%. With no material change in the severity of the key variable impacting consumer credit performance, the resulting projected losses for the credit card portfolio were correspondingly stable.

Conversely, the projected loss rate for the **Commercial Real Estate** portfolio increased materially by 2.6 percentage points. This increase is a direct consequence of the heightened stress applied to the commercial property market in the 2026 scenario. While the peak unemployment rate was unchanged YoY, the 2026 scenario introduced a substantially more severe shock to CRE prices, projecting a trough decline of 39% compared to 30% in the 2025 scenario. This 9 percentage point greater decline in asset values is the primary factor driving the significant increase in projected losses for Citigroup's CRE portfolio.

[Usage] Prompt Tokens: 168,661 | Output Tokens: 2,460
[Cost] Input Cost: $0.210826 | Output Cost: $0.024600 | Total Cost: $0.235426
→ Macro numbers found. Running DeepEval Economic Reasoning Check...
→ DeepEval Score: 90.0% | Reason: The response strongly aligns with the evaluation steps. It accurately identifies the changes in macroeconomic severity (CRE price trough worsening from -30% to -39%, unemployment stable at 10.0%) and references these consistently in the executive summary. The causal reasoning is sound: credit card loss stability is logically tied to unchanged unemployment, and the CRE loss increase (+2.6 p.p.) is directly linked to the 9 p.p. greater CRE price decline. Domain knowledge is correctly applied (unemployment as the driver for unsecured consumer credit, CRE price index and unemployment for commercial real estate) without contradicting the input. The synthesis is coherent with no notable logical gaps. Minor caveat: the summary slightly glosses over the small -0.3 p.p. credit card decrease by describing it as 'stability,' though it does not misrepresent the data.
→ DeepEval Usage: Input 2,863 / Output 569 | Total Cost: $0.028540
✅ Validation: PASSED

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🏃 RUNNING COMPLEX PROMPT | Model: claude-sonnet-4-6
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[Output]
Macroeconomic Attribution Analysis: Citigroup Credit Cards & CRE Portfolios (2025 vs. 2026 CCAR)

Section 1: Loss Rate Change

Extracted directly from the Results documents (Table 10 in 2025 Results; Table 10 in 2026 Results):

Portfolio	2025 Projected Loss Rate	2026 Projected Loss Rate	YoY Change (2026 minus 2025)
Credit Cards	16.6%	16.3%	**-0.3 pp**
Commercial Real Estate, Domestic	8.3%	10.9%	**+2.6 pp**

Sources:
- 2025 CCAR Results, Table 10 (Citigroup row): Credit Cards = 16.6%; CRE Domestic = 8.3%
- 2026 CCAR Results, Table 10 (Citigroup row): Credit Cards = 16.3%; CRE Domestic = 10.9%

Section 2: Macroeconomic Drivers

Credit Card Portfolio

The Fed documents note that "some loan portfolios are sensitive to home prices or unemployment rates and may experience high stressed loss rates due to the considerable stress on these factors in the severely adverse scenario" (2025 Results, p. 21). Credit card losses are driven by consumer financial stress. Based on the documents' explicit commentary and standard financial domain knowledge, the primary macroeconomic drivers for credit card loss rates are:

1. **Unemployment Rate** — The single most direct driver of consumer credit deterioration. Rising unemployment reduces household income, impairing debt service capacity and driving charge-offs. The Fed's scenario design framework explicitly identifies unemployment as a primary scenario variable, and the documents repeatedly cite unemployment as a key driver of consumer loan losses.
2. **Real GDP Growth (Peak-to-Trough Decline)** — Broad economic contraction reduces consumer income and spending capacity, amplifying credit card delinquencies.
3. **Equity Prices** — Declining equity prices reduce household wealth, tightening consumer balance sheets and increasing default propensity, particularly for higher-income cardholders.

Commercial Real Estate (CRE) Domestic Portfolio

The Fed documents explicitly state that "declines in commercial real estate prices should be assumed to be concentrated in properties most at risk of a sustained drop in income and asset values" (2025 Scenarios, p. 7). The primary macroeconomic drivers for CRE loss rates are:

1. **Commercial Real Estate (CRE) Price Decline** — The most direct driver. Collateral value erosion increases loss-given-default (LGD) on CRE loans. The Fed's scenario design framework explicitly includes CRE prices as a key scenario variable.
2. **Unemployment Rate** — Rising unemployment reduces occupancy rates and rental income for commercial properties, impairing debt service coverage ratios (DSCR) and increasing probability of default (PD).
3. **BBB Corporate Bond Spread** — Widening credit spreads reflect deteriorating corporate credit conditions, which directly impair the ability of commercial tenants and CRE borrowers to service debt, and increase capitalization rates, further depressing property values.
4. **10-Year Treasury Yield (Trough Level)** — Long-term interest rates affect CRE capitalization rates and refinancing risk. In the 2026 scenario, the notably higher trough for the 10-year Treasury (2.3% vs. 1.0% in 2025) means interest rates remain elevated throughout the stress horizon, increasing refinancing costs for CRE borrowers and suppressing property valuations relative to the 2025 scenario.

Section 3: Macroeconomic Variable Analysis

All values are extracted from the Severely Adverse Scenario domestic variable tables (Table 4.A) in the 2025 and 2026 Stress Test Scenarios documents.

Methodology Notes:
- **Unemployment Rate**: Peak value reached during the scenario horizon.
- **Real GDP**: Peak-to-trough percentage decline (as reported in the scenario comparison tables within the Results documents).
- **CRE Price Decline**: Peak-to-trough percentage decline from jump-off value (as reported in scenario comparison tables and scenario narrative).
- **Equity Price Decline**: Peak-to-trough percentage decline from jump-off value (as reported in scenario comparison tables).
- **BBB Corporate Yield (Peak Spread)**: The peak BBB yield level reached in the scenario, which reflects the maximum credit stress. The spread over the 10-year Treasury at that point is the relevant severity measure.
- **10-Year Treasury Yield (Trough)**: The lowest level reached, representing the maximum interest rate relief (or in 2026, the constrained relief).

Macro Variable	2025 Severity	2026 Severity	YoY Difference
Unemployment Rate (Peak)	10.0% (Q3 2026)	10.0% (Q3 2027)	**0.0 pp** (same peak, later timing)
Unemployment Rate (Increase from Jump-Off)	+5.9 pp (from 4.1%)	+5.5 pp (from 4.5%)	**-0.4 pp** (smaller rise in 2026)
Real GDP (Peak-to-Trough Decline)	-7.8%	-4.6%	**+3.2 pp less severe in 2026**
CRE Price Decline (Peak-to-Trough)	-30%	-39%	**-9.0 pp (more severe in 2026)**
Equity Price Decline (Peak-to-Trough)	-50%	-58%	**-8.0 pp (more severe in 2026)**
BBB Corporate Yield (Peak Level)	6.0% (Q3 2025)	8.2% (Q2 2026)	**+2.2 pp (more severe in 2026)**
BBB Corporate Spread over 10-yr Treasury at Peak	~5.0 pp (6.0% BBB - 1.0% 10yr)	~5.9 pp (8.2% BBB - 2.3% 10yr)	**+0.9 pp (more severe in 2026)**
10-Year Treasury Yield (Trough)	1.0% (Q2 2025)	2.3% (Q4 2026)	**+1.3 pp (higher floor = less rate relief in 2026)**
3-Month Treasury Rate (Trough)	0.1%	0.1%	**0.0 pp**
House Price Decline (Peak-to-Trough)	-33%	-30%	**+3.0 pp less severe in 2026**

Data Sources:
- 2025 Scenarios, Table 4.A: BBB corporate yield peaks at 6.0% in Q3 2025; 10-year Treasury troughs at 1.0% in Q2 2025; unemployment peaks at 10.0% in Q3 2026.
- 2026 Scenarios, Table 4.A: BBB corporate yield peaks at 8.2% in Q2 2026; 10-year Treasury troughs at 2.3% in Q4 2026; unemployment peaks at 10.0% in Q3 2027.
- Scenario comparison tables in both Results documents confirm GDP, CRE, house price, and equity decline magnitudes.

Section 4: Executive Summary

CCAR 2025 vs. 2026 — Citigroup Credit Card & CRE Loss Rate Attribution

Overview

The 2026 CCAR severely adverse scenario produced materially divergent outcomes for Citigroup's two analyzed portfolios relative to 2025. Citigroup's credit card loss rate declined modestly by **0.3 percentage points** (from 16.6% to 16.3%), while its CRE domestic loss rate surged by **2.6 percentage points** (from 8.3% to 10.9%). These opposing movements are not contradictory — they are the direct and logical consequence of a scenario that was specifically calibrated to be *less severe* for consumer credit stress but *substantially more severe* for commercial real estate and corporate credit conditions.

Credit Cards: Modest Improvement Driven by Reduced Consumer Stress

The marginal improvement in Citigroup's credit card loss rate from 16.6% to 16.3% is attributable to a modestly less severe consumer stress environment in the 2026 scenario, despite the same peak unemployment level.

The **unemployment rate** reached an identical peak of 10.0% in both scenarios. However, the *path* to that peak differs in a manner that matters for credit card loss modeling. In 2025, unemployment rose **5.9 percentage points** from a jump-off of 4.1%, while in 2026 it rose only **5.5 percentage points** from a higher jump-off of 4.5%. Critically, the 2026 scenario reaches its unemployment peak in Q3 2027 — the **seventh quarter** of the nine-quarter horizon — compared to Q3 2026 in the 2025 scenario. This later timing means the most severe unemployment stress is compressed into fewer remaining quarters of the projection horizon, reducing the cumulative consumer credit losses recognized over the full nine-quarter window.

Furthermore, **real GDP's peak-to-trough decline** was substantially less severe in 2026 (-4.6%) compared to 2025 (-7.8%), a difference of 3.2 percentage points. This shallower economic contraction implies less aggregate income destruction for households, supporting debt service capacity and moderating credit card charge-off rates. The Fed's own decomposition in the 2026 Results document (Figure 2) confirms that loan loss provisions increased year-over-year in aggregate, but this was driven by loan balance growth (credit cards grew ~10% in 2025) rather than by higher loss *rates* on the consumer side. For Citigroup specifically, the slightly lower loss rate reflects the scenario's reduced consumer-side severity, partially offset by Citigroup's large international consumer card exposure (18.8% other consumer loss rate in 2025 vs. 21.5% in 2026, suggesting international consumer stress increased).

The **equity price decline** deepened from -50% to -58%, which would ordinarily pressure consumer balance sheets. However, this variable has a secondary effect on credit card losses relative to unemployment and income, and its incremental impact was insufficient to reverse the directional improvement driven by the shallower GDP contraction and the later-arriving unemployment peak.

Commercial Real Estate: Significant Deterioration Driven by Three Compounding Forces

The 2.6 percentage point increase in Citigroup's CRE domestic loss rate — from 8.3% to 10.9% — reflects the convergence of three macroeconomic forces that are each individually more severe in the 2026 scenario and that interact multiplicatively in CRE credit modeling.

Force 1: Dramatically Deeper CRE Price Decline (+9 pp more severe)

The most direct driver of CRE loss rates is collateral value. The 2026 scenario features a **39% peak-to-trough decline** in commercial real estate prices, compared to only **30%** in 2025 — a 9 percentage point increase in severity. The Fed's 2026 Scenarios document explicitly explains this calibration: CRE prices were set "near the top one-third of their ranges of severity" given that equity prices had risen 16% in 2025 and the Board sought to maintain appropriate overall scenario severity. This deeper collateral erosion directly increases loss-given-default (LGD) across Citigroup's CRE portfolio. When a borrower defaults on a CRE loan, the bank's recovery depends on the liquidation value of the underlying property. A 39% price decline versus a 30% decline translates mechanically into higher net losses per defaulted loan, driving the loss rate higher even if the probability of default were held constant.

Force 2: Elevated and Persistent Interest Rates — The Refinancing Trap

The 2026 scenario introduces a qualitatively different interest rate environment that is particularly punishing for CRE borrowers. The **10-year Treasury yield troughs at 2.3%** in the 2026 scenario versus only **1.0%** in 2025 — a 1.3 percentage point higher floor. This is not a minor technical difference; it represents a fundamentally different stress narrative. In 2025, the scenario featured a sharp flight-to-safety rally that drove long rates to near-zero, providing meaningful refinancing relief for CRE borrowers with maturing debt. In 2026, the scenario is characterized by an inflationary shock (the global market shock features "higher inflation expectations"), meaning long-term rates remain elevated even as the economy contracts. This "stagflationary" dynamic is the worst possible environment for CRE: property income falls as vacancies rise and rents decline, while refinancing costs remain high because long-term rates do not fall commensurately. CRE loans with near-term maturities face a "refinancing wall" — they cannot roll over at lower rates, and their collateral values have simultaneously declined

39%. This dynamic is a primary driver of the elevated CRE loss rate in 2026.

Force 3: Wider Corporate Credit Spreads Amplifying Tenant and Borrower Stress

The **BBB corporate yield peaked at 8.2%** in Q2 2026 under the 2026 scenario, compared to a peak of **6.0%** in Q3 2025 under the 2025 scenario — a 2.2 percentage point increase in peak credit stress. The implied BBB spread over the 10-year Treasury at the peak was approximately 5.9 percentage points in 2026 versus approximately 5.0 percentage points in 2025. Wider corporate spreads reflect more severe corporate credit deterioration, which impairs CRE performance through two channels: (1) commercial tenants facing higher borrowing costs and tighter credit conditions are more likely to default on leases, reducing property cash flows and impairing debt service coverage ratios; and (2) CRE borrowers themselves — many of whom are corporate entities — face higher refinancing costs and reduced access to capital markets, increasing their probability of default. The 2026 scenario's more severe corporate credit stress therefore amplifies both the PD and LGD components of Citigroup's CRE loss rate.

Synthesis and Conclusion

The divergence between Citigroup's credit card and CRE loss rate trajectories from 2025 to 2026 is a textbook illustration of how scenario design choices create portfolio-specific outcomes. The 2026 scenario was deliberately calibrated — consistent with the Fed's Proposed 2025 Scenario Design Policy Statement — to reflect a world where equity prices had risen significantly (necessitating a deeper equity shock of -58%), CRE prices had stabilized at elevated levels (necessitating a deeper CRE shock of -39%), and the inflationary environment of 2025 meant that long-term interest rates could not fall as sharply in a stress scenario (constraining the 10-year Treasury trough at 2.3%).

For **credit cards**, the scenario's shallower GDP contraction (-4.6% vs. -7.8%) and the later timing of peak unemployment provided modest relief, resulting in a -0.3 pp improvement in Citigroup's loss rate. For **CRE**, the trifecta of a 9 pp deeper collateral price decline, a 1.3 pp higher interest rate floor, and a 2.2 pp wider peak corporate credit spread created compounding adverse pressure, driving a +2.6 pp deterioration in Citigroup's loss rate. The net result is that Citigroup's CRE portfolio, at a 10.9% loss rate, now represents a materially elevated risk concentration under the 2026 stress framework — one that risk management should monitor closely given the real-world parallels to the scenario's stagflationary narrative, including persistent inflation, elevated long-term rates, and ongoing commercial real estate price pressure in office and retail segments.

[Usage] Prompt Tokens: 165,235 | Output Tokens: 3,869
[Cost] Input Cost: $0.495705 | Output Cost: $0.058035 | Total Cost: $0.553740

→ Macro numbers found. Running DeepEval Economic Reasoning Check...
→ DeepEval Score: 90.0% | Reason: The response establishes a clear, logical connection between macroeconomic severity changes and loan loss rate movements. It correctly identifies the divergence: credit cards improved (-0.3 pp) due to shallower GDP contraction (-4.6% vs -7.8%) and later-timed unemployment peak, while CRE deteriorated (+2.6 pp) due to deeper CRE price decline (-39% vs -30%), higher Treasury trough (2.3% vs 1.0%), and wider BBB spreads. The economic reasoning linking macro variables to portfolios is accurate and domain-appropriate (collateral value/LGD for CRE, unemployment/income for credit cards, refinancing-wall dynamics from elevated rates). The directional consistency is strong—more severe CRE-related variables map to higher CRE losses, and less severe consumer variables map to modestly lower card losses. Minor shortcomings: the explanation of a same-peak unemployment (10.0%) yielding lower card losses relies on nuanced timing/path arguments that, while plausible, are somewhat speculative, and some data values (e.g., exact table figures) cannot be independently verified from the summary alone. Overall, the synthesis is comprehensive, internally consistent, and well-aligned with the implied data relationships.
→ DeepEval Usage: Input 7,100 / Output 672 | Total Cost: $0.052300

✅ Validation: PASSED